

Student Database Management System Final Report by NRS Company

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COSC-3336: Intro to Database Systems

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April 28, 2024

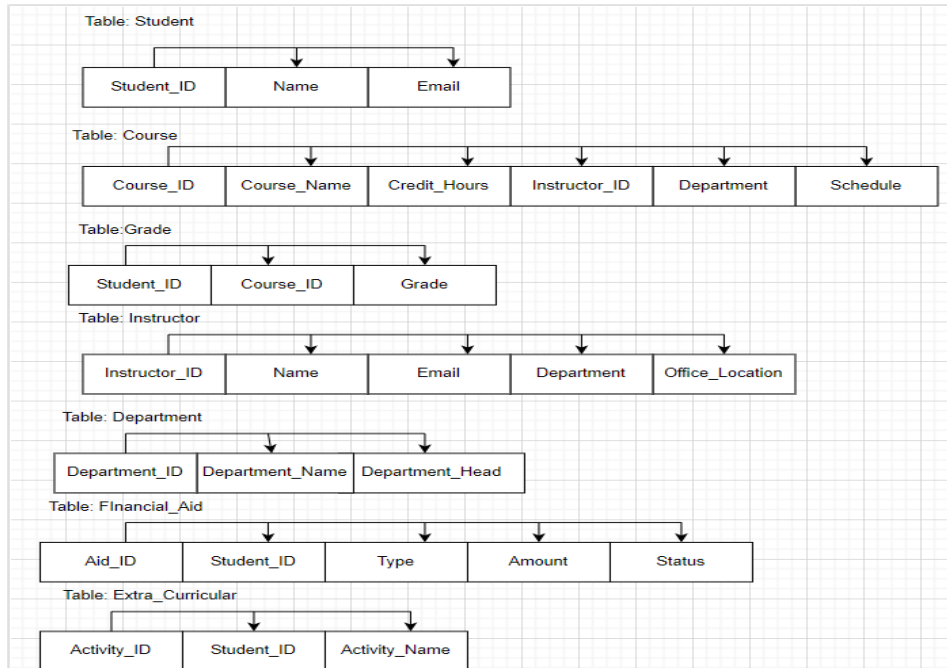
Phase #1 Report

During phase one of our project, our group decided our database would be based around a student database. We made this decision because we figured that it was an idea that we will all have knowledge about and that this project would be fun, scalable, user friendly and efficient. We also believe we have seen a decent number of student databases in our lives, so we would have many ideas to design our own. This is also part of the project where we assigned roles for each person in our group. We decided that Hunter Nichols would be the Requirements Manager, which meant that he makes sure that the database meets every requirement for the project. Rhett Richardson would be the Quality Manager, making sure we deliver the highest quality database at our current skill level. Derrick Subnaik was assigned to be our Project Manager; he was responsible for keeping the team on task and alerting them when deadlines needed to be met. Overall, this phase was our planning phase as we were able to come together and set the basics so we could work effectively on the project together.

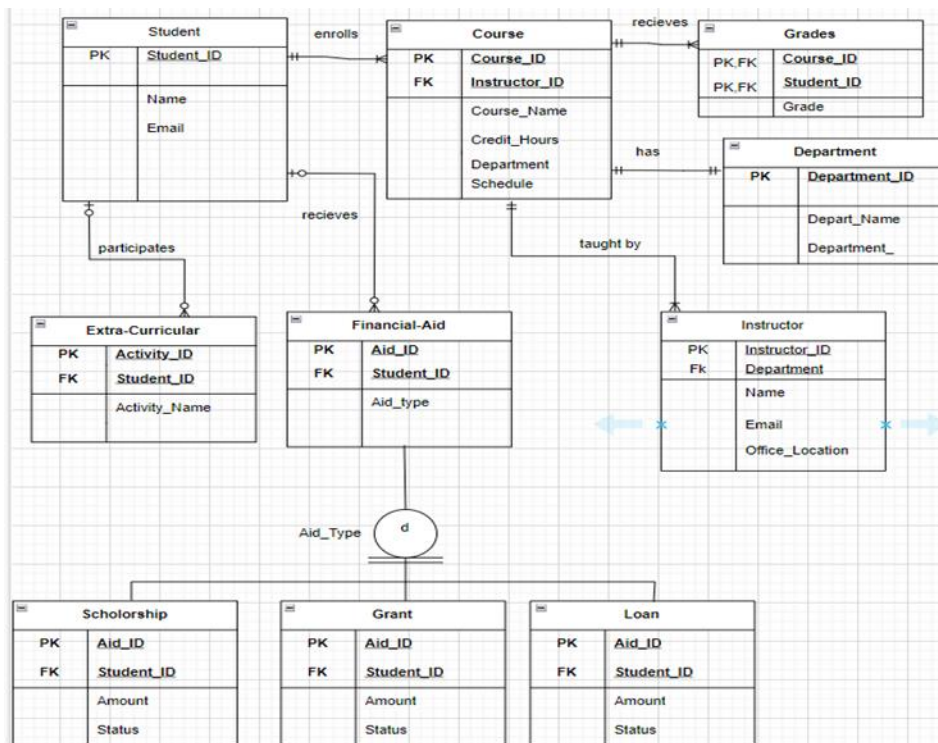
Phase #2 Report

During phase two of our project, we developed what would be within the database. Our goal in the creation of this Database System is to create a system that will allow for efficient storage, retrieval, and management of students at Academic Institutions. For our system we had seven entities, Student, Course, Enrollment, Instructor, Department, Financial-Aid, and Extracurricular. We decided that we wanted to produce our product to store anything from Course grades to student extracurriculars. In this phase we also Normalized our database table, and our Entity Relation Diagram. The last thing we did for this phase was enter 50 using SQL to start creating our database. Overall, this phase was preplanning what we wanted within our database and what we thought would be most beneficial to the user.

Normalization of Tables



ERD



Phase #3 Report

During Phase 3 of our project, we finished implementing our database in SQL. We created our seven tables and conducted testing to ensure all entries were entered correctly. After this we conducted the following SQL queries that could be scenarios for a student database Academic Institutions.

```
SELECT * FROM Financial_Aid WHERE Aid_Type = 'Scholarship' AND Aid_Amount BETWEEN 4500 AND 5500;
```

- This query selects all records from the Financial_Aid table where Aid_Type is 'Scholarship' and Aid_Amount is between 4500 and 5500

Aid_ID	Student_ID	Aid_Type	Aid_Amount	Aid_Status
S1	1	Scholarship	5000.00	Approved
S10	28	Scholarship	5400.00	Approved
S11	31	Scholarship	5300.00	Approved
S12	34	Scholarship	4500.00	Approved
S3	7	Scholarship	5500.00	Approved
S4	10	Scholarship	4800.00	Approved
S5	13	Scholarship	5100.00	Approved
S6	16	Scholarship	4700.00	Approved
S7	19	Scholarship	4900.00	Approved
S8	22	Scholarship	5200.00	Approved
S9	25	Scholarship	4600.00	Approved

```
SELECT * FROM Student WHERE Aid_ID LIKE 'L%';
```

- This query selects all data of students who do not receive a loan

ent_ID	Student_name	Student_email	Aid_ID
3	Derrick Subnaik	dsubnaik@islander.tamucc.edu	L1
6	Peyton Lewis	plewis@islander.tamucc.edu	L2
9	Jenna Martin	jmartin@islander.tamucc.edu	L3
12	Rey Gonzalez	rgonzalez@islander.tamucc.edu	L4
15	Blessing Rain	brain@islander.tamucc.edu	L5
18	Cecilia Padilla	cpadilla@islander.tamucc.edu	L6
21	Bobby Lewis	blewis@islander.tamucc.edu	L7
24	Karley Lewis	klewis@islander.tamucc.edu	L8
27	Jane Smith	jsmith@islander.tamucc.edu	L9
30	Harrison Nguyen	hnguyen@islander.tamucc.edu	L10
33	Savannah Carter	scarter@islander.tamucc.edu	L11
36	Lucas Khan	lkhan@islander.tamucc.edu	L12
39	Mia Campbell	mcampbell@islander.tamucc.edu	L13
42	Sebastian Lee	slee@islander.tamucc.edu	L14
44	Nova Taylor	ntaylor@islander.tamucc.edu	L15
47	Caleb Clark	cclark@islander.tamucc.edu	L16

```
SELECT Course_ID, Course_Name FROM Course WHERE Credit_Hours >= 4;
```

- This query selects the Course_ID and Course_Name of courses where Credit_Hours are greater than or equal to 4

Course_ID	Course_Name
COSC 1435	Intro to Problem Solving with Computers 1
COSC 1436	Intro to Problem Solving with Computers 2
COSC 2437	Data Structures
MATH 2413	Calculus 1
MATH 2414	Calculus 2

```
SELECT Aid_Type, COUNT(*) AS Aid_Count, SUM(Aid_Amount) AS Total_Aid_Amount FROM
Financial_Aid WHERE Aid_Status = 'Approved' GROUP BY Aid_Type ORDER BY Aid_Type;
```

- This query selects records from the Financial_Aid table where Aid_Amount is greater than 5000 and Aid_Type is 'Loan'

Aid_Type	Aid_Count	Total_Aid_Amount
Grant	4	12000.00
Loan	7	49900.00
Scholarship	17	85100.00

```
SELECT COUNT(*) AS Approved_Aid_Count FROM Financial_Aid WHERE Aid_Status = 'Approved';
```

- This query calculates the count of each Aid_Type and the total amount of aid for each type from the Financial_Aid table where Aid_Status is 'Approved'

Approved_Aid_Count
28

```
SELECT SUM(Aid_Amount) AS Total_Scholarship_Aid FROM Financial_Aid WHERE Aid_Type =
'Scholarship';
```

- This query calculates the total amount of scholarship aid from the Financial_Aid table where Aid_Type is 'Scholarship'

Total_Scholarship_Aid
85100.00

```
SELECT AVG(Aid_Amount) AS Average_Loan_Aid FROM Financial_Aid WHERE Aid_Type = 'Loan';
```

- This query calculates the average amount of loan aid from the Financial_Aid table where
Aid_Type is 'Loan'

Average_Loan_Aid
7406.250000

```
SELECT MAX(Aid_Amount) AS Max_Pending_Aid FROM Financial_Aid WHERE Aid_Status = 'Pending';
```

- This query calculates the maximum pending aid amount from the Financial_Aid table where
Aid_Status is 'Pending'

Max_Pending_Aid
8200.00

```
SELECT MIN(Aid_Amount) AS Min_Denied_Aid FROM Financial_Aid WHERE Aid_Status = 'Denied';
```

- This query calculates the minimum denied aid amount from the Financial_Aid table where
Aid_Status is 'Denied'

Min_Denied_Aid
1700.00

```
SELECT Aid_ID, Student_ID, Aid_Amount, Aid_Type, Aid_Status FROM Financial_Aid WHERE Aid_Status = 'Approved' ORDER BY Aid_Amount DESC;
```

- This query selects records from the Financial_Aid table where Aid_Status is 'Approved' and orders them by Aid_Amount in descending order

Aid_ID	Student_ID	Aid_Amount	Aid_Type	Aid_Status
L1	3	8000.00	Loan	Approved
L9	27	7400.00	Loan	Approved
L7	21	7300.00	Loan	Approved
L5	15	7200.00	Loan	Approved
L12	36	7100.00	Loan	Approved
L14	42	6900.00	Loan	Approved
L2	6	6000.00	Loan	Approved
S17	49	5900.00	Scholarship	Approved
S15	43	5800.00	Scholarship	Approved
S13	37	5700.00	Scholarship	Approved
S3	7	5500.00	Scholarship	Approved
S10	28	5400.00	Scholarship	Approved
S11	31	5300.00	Scholarship	Approved
S8	22	5200.00	Scholarship	Approved
S5	13	5100.00	Scholarship	Approved
S1	1	5000.00	Scholarship	Approved
S7	19	4900.00	Scholarship	Approved
S4	10	4800.00	Scholarship	Approved
S6	16	4700.00	Scholarship	Approved
S9	25	4600.00	Scholarship	Approved
S12	34	4500.00	Scholarship	Approved

SELECT Student.Student_ID, Student.Student_name, Extra_Curricular.Activity_Name FROM Student

INNER JOIN Extra_Curricular ON Student.Student_ID = Extra_Curricular.Student_ID;

- This query selects Student_ID, Student_name, and Activity_Name from the Student and Extra_Curricular tables using an INNER JOIN on Student_ID

Student_ID	Student_name	Activity_Name
1	Hunter Nichols	Chess Club
2	Rhett Richardson	Debate Society
3	Derrick Subnaik	Music Band
4	Daniel Carlos	Drama Club
5	Boris Hernandez	Art Club
9	Jenna Martin	Music Band
10	Angela Franco	Drama Club
11	Kayla Greir-Porter	Basketball Team
12	Rey Gonzalez	Debate Society
13	Sarah Decker	Music Band
14	Maria Paula Islas	Debate Society
15	Blessing Rain	Music Band
16	Natalie Johnson	Drama Club
21	Bobby Lewis	Music Band
22	John Cummings	Debate Society
23	Keaton Grace	Music Band
24	Karley Lewis	Debate Society
25	Stone Groves	Music Band
26	John Doe	Drama Club
29	Aurora Blake	Drama Club
30	Harrison Nguyen	Art Club
31	Penelope Patel	Music Band
32	Leo Ramirez	Debate Society
33	Savannah Carter	Music Band
34	Oliver Thompson	Debate Society
35	Isabella Davis	Music Band
36	Lucas Khan	Art Club
39	Mia Campbell	Music Band
40	Elijah Brown	Drama Club
41	Layla Lewis	Basketball Team
42	Sebastian Lee	Debate Society
43	Aiden Johnson	Music Band
44	Nova Taylor	Art Club
45	Mason White	Music Band
46	Ava Adams	Art Club
49	Ethan Wilson	Music Band
50	Jasper Watson	Drama Club

SELECT Student.Student_ID, Student.Student_name, Enrollment.Course_ID, Enrollment.Grade FROM

Student LEFT JOIN Enrollment ON Student.Student_ID = Enrollment.Student_ID;

- This query selects Student_ID, Student_name, Course_ID, and Grade from the Student and Enrollment tables using a LEFT JOIN on Student_ID

- Screenshots are not provided because it would take too many to fit on this page

```
SELECT Course.Course_ID, Course.Course_Name, Instructor.Instructor_Name FROM Course RIGHT JOIN
Instructor ON Course.Instructor_ID = Instructor.Instructor_ID;
```

- This query selects Course_ID, Course_Name, and Instructor_Name from the Course and Instructor tables using a RIGHT JOIN on Instructor_ID

Course_ID	Course_Name	Instructor_Name
		Lauren Friley
COMM 1301	Foundation of Communication	John Harrison
MATH 2414	Calculus 2	John Harrison
MATH 2413	Calculus 1	Anna Andrei
PHIL 1301	Introduction to Philosophy	David Fuentes
ARTS 1301	Art and Society	Habib Senol
HIST 1302	US History since 1865	Habib Senol
HIST 1301	US History to 1865	Susanaa Rijkoof
POLS 2306	State and Local Government	Susanaa Rijkoof
POLS 2305	US Government and Politics	Nene Coulibaly
MATH 1416	Trigonometry	Mamta Yadav
COSC 2334	Computer Architecture	Mamta Yadav
COSC 1435	Intro to Problem Solving with Computers 1	Huyen Le
COSC 2437	Data Structures	Huyen Le
COSC 1436	Intro to Problem Solving with Computers 2	Kadir Demir
COSC 3370	Software Engineering	Kadir Demir
COSC 3324	Object-Oriented Programming	Dulal Kar
COSC 3346	Operating Systems	Mahboobeh Haghpourast
COSC 3336	Intro to Database Systems	Christopher Andrews
ENGL 3310	Technical and Professional Writing	Yu Fan
MATH 2305	Discrete Mathematics	Harry Lee
MATH 3342	Applied Probability and Statistics	

```
SELECT Course.Course_ID, Course.Course_Name, Department.Department_Name FROM Course FULL
OUTER JOIN Department ON Course.Department_ID = Department.Department_ID WHERE
Course.Credit_Hours > 3;
```

- This query selects Course_ID, Course_Name, and Department_Name from the Course and Department tables using a FULL OUTER JOIN on Department_ID, and filters results where Credit_Hours are greater than 3

- Screenshots are not provided because it would take too many to fit on this page

```
SELECT T1.Course_ID, T2.Course_ID FROM Course T1, Course T2 WHERE T1.Course_ID < T2.Course_ID;
```

- This query selects pairs of Course_IDs from the Course table where the first Course_ID is less than the second Course_ID
- Screenshots are not provided because it would take too many to fit on this page

```
SELECT Student.Student_ID, Student.Student_name, Course.Course_ID, Course.Course_Name FROM
Student CROSS JOIN Course;
```

- This query selects Student_ID, Student_name, Course_ID, and Course_Name using a CROSS JOIN between the Student and Course tables
- Screenshots are not provided because it would take too many to fit on this page

SELECT Instructor_Name, Instructor_Email FROM Instructor WHERE Department_ID IN ('ARTS', 'MD');

- This query selects Instructor_Name and Instructor_Email from the Instructor table where the Department_ID is either 'ARTS' or 'MD'

Instructor_Name	Instructor_Email
John Harrison	jharrison@islander.tamucc.edu
Anna Andrei	aandrei@islander.tamucc.edu
David Fuentes	dfuentes@islander.tamucc.edu
Nene Coulibaly	ncoulibaly@islander.tamucc.edu
Yu Fan	yfan@islander.tamucc.edu
Harry Lee	hlee@islander.tamucc.edu

We also did the following queries when it came to updating and altering table.

ALTER TABLE Student ADD Student_Gender VARCHAR(10);

- This query adds a new column called Student_Gender to the Student table with a data type of VARCHAR(10)

ALTER TABLE Student DROP COLUMN Student_Gender;

- This query removes the column Student_Gender from the Student table

UPDATE Course SET Credit_Hours = 5 WHERE Course_ID = 'MATH 2413';

- This query updates the Credit_Hours column in the Course table to 5 where the Course_ID is 'MATH 2413'

DELETE FROM Enrollment WHERE Student_ID = 30 AND Course_ID = 'COSC 3370';

- This query deletes a record from the Enrollment table where Student_ID is 30 and Course_ID is 'COSC 3370'

References

Online course schedule. Texas A&M University-Corpus Christi. (n.d.).
<https://banner.tamucc.edu/schedule/>